



WHAT THE





- Mickey and Jesse
- Who are we



A long time ago in a computer far, far away...

- I had a web cam



A long time ago in a computer far, far away...

- Funny Story...
 - One day my webcam was flaky
 - Firmware update?
 - Download update package!

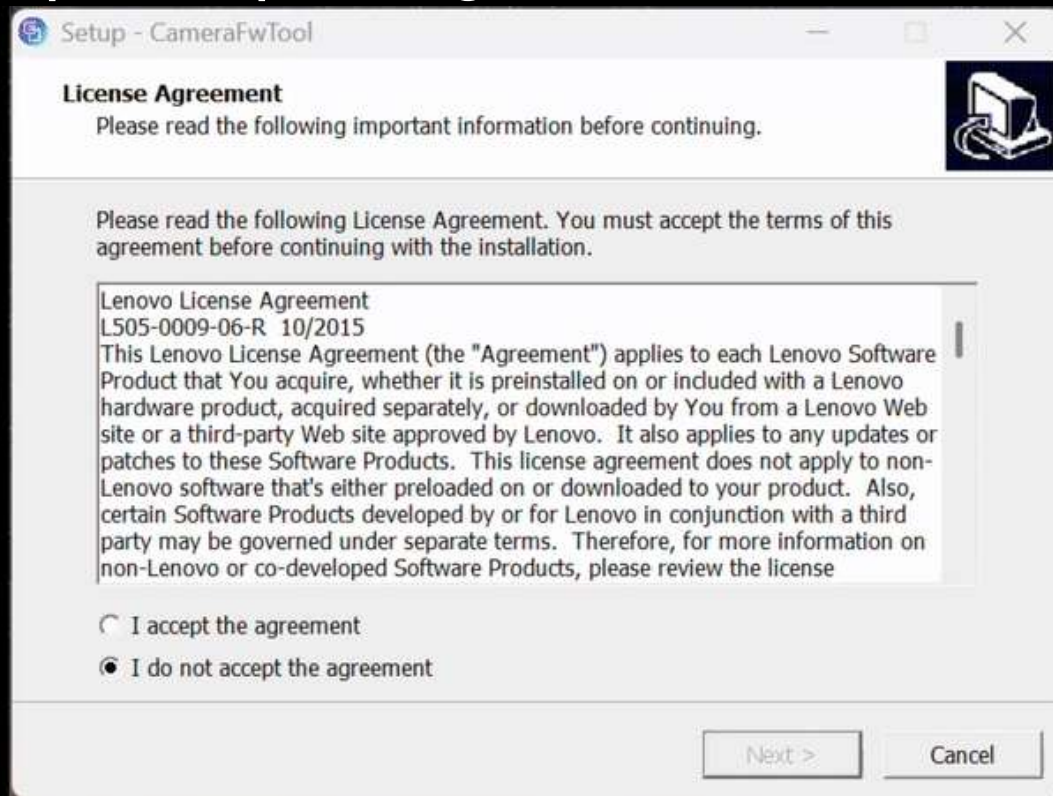


camerafwtool_hd510_1_0_19_0525_v4_6_2.exe



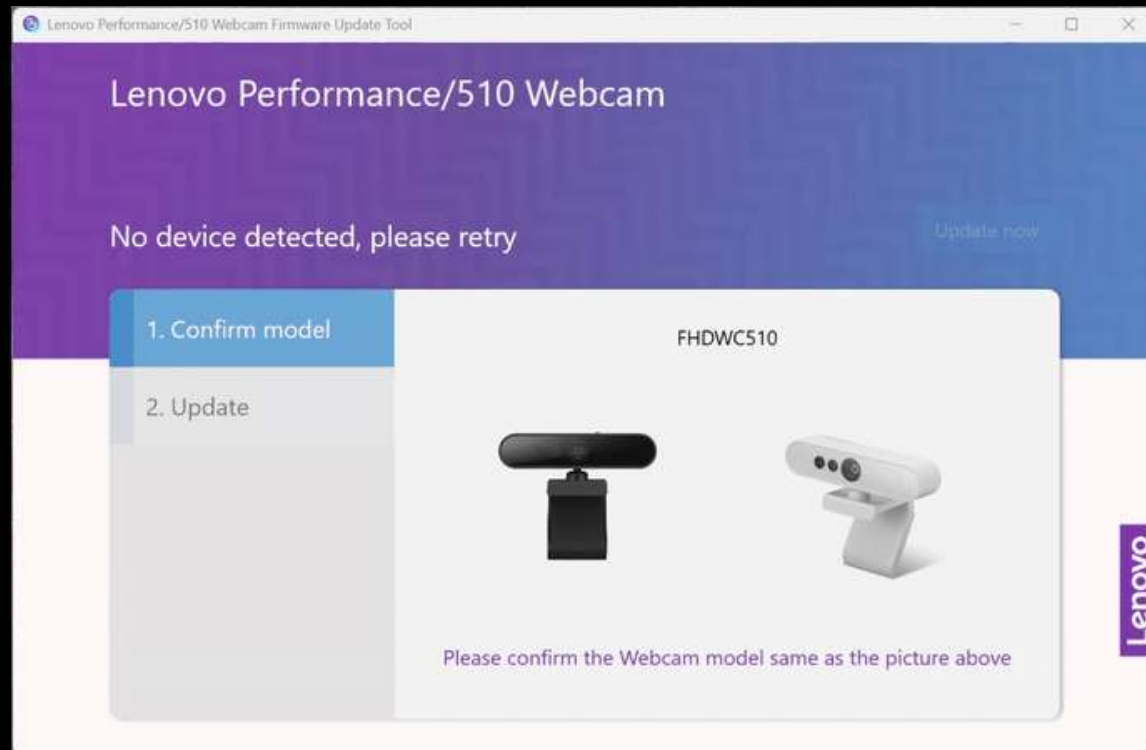
Background

- Install update package!



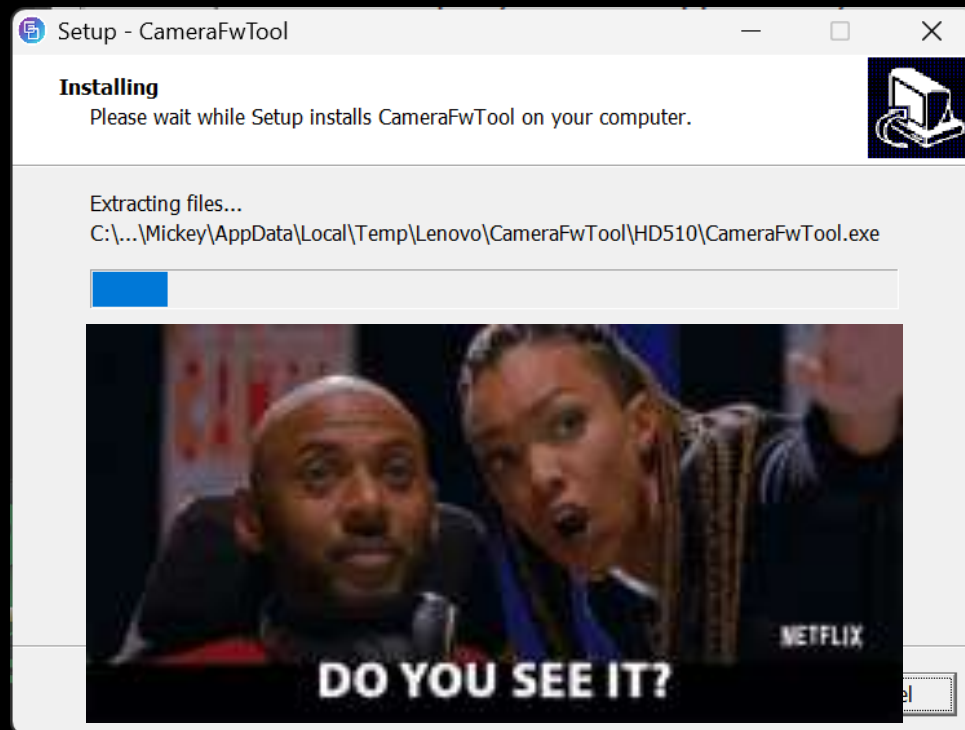
Background

- Install update package!



Background

- Install update package!



Background

- Let's open that folder

Name	Size
Resource	
[[ota	1 KB
AitUVCExtApi.dll	195 KB
auto_update.txt	1 KB
CameraFwTool.exe	1,477 KB
CameraFwTool.exe.config	2 KB
dummy.bin	4 KB
HD510Sdk.dll	2,084 KB
lenovo_hd510_ota_v4.6.2.bin	7,808 KB
log4net.dll	275 KB
make_imge.sh	3 KB
u-boot.bin	344 KB
uboot_s.bin	100 KB
usb_updater.bin	64 KB



Background

- Can you see anything interesting?



Background

- Can you see anything interesting?

 CameraFwTool.exe.config	2 KB
 dummy.bin	4 KB
 HD510Sdk.dll	2,084 KB
 lenovo_hd510_ota_v4.6.2.bin	7,808 KB



Background

lenovo_hd510_ota_v4.6.2.bin

7,808 KB



Background

- Ok, let's binwalk a little

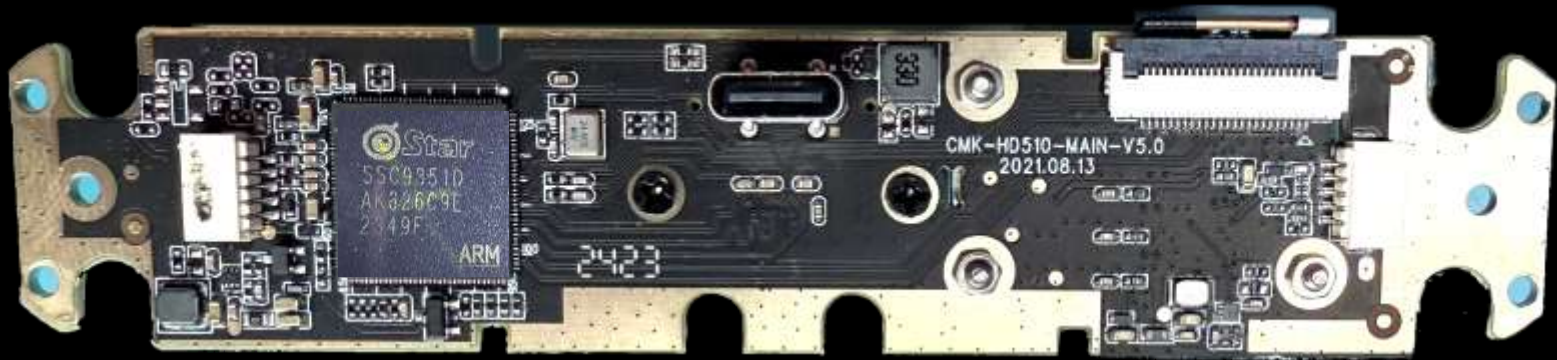
DECIMAL	HEXADECIMAL	DESCRIPTION
1835008	0x1C0000	gzip compressed data, operating system: Unix, timestamp: 1970-01-01 00:00:00, total size: 2700525 bytes
4653056	0x470000	SquashFS file system, little endian, version: 4.0, compression: xz, inode count: 26, block size: 131072, image size: 2654642 bytes, created: 2022-05-21 07:25:29
7471104	0x720000	JFFS2 filesystem, little endian, nodes: 22, total size: 327692 bytes

- Now we REALLY need to open this thing up!

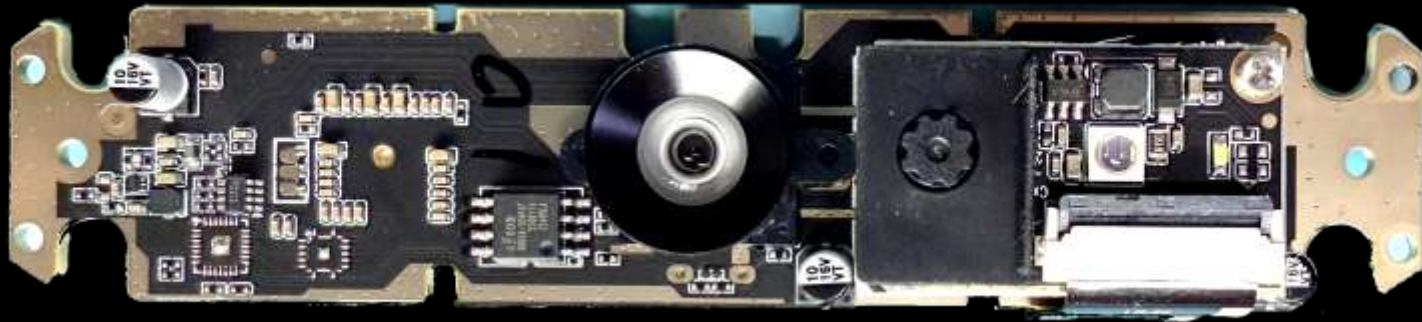


Background

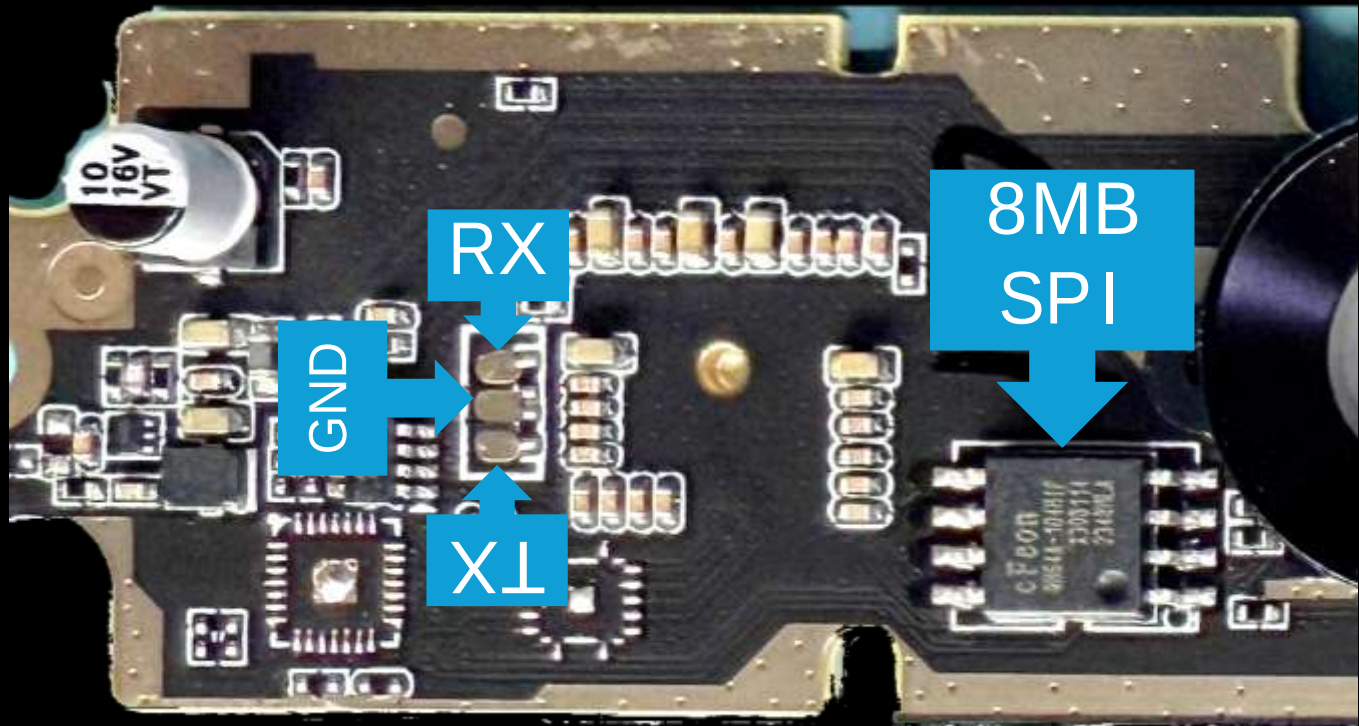
- BACK



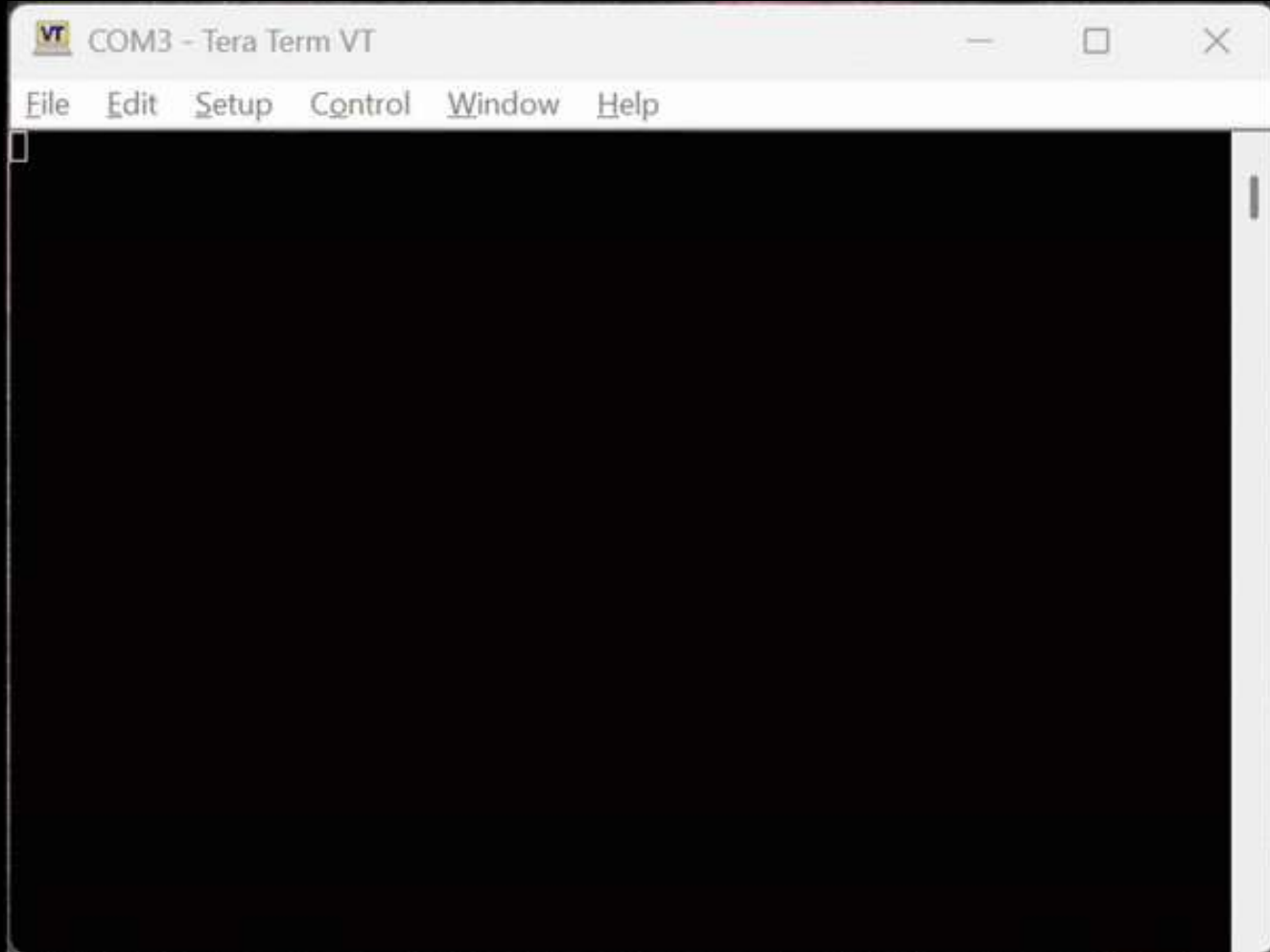
- FRONT

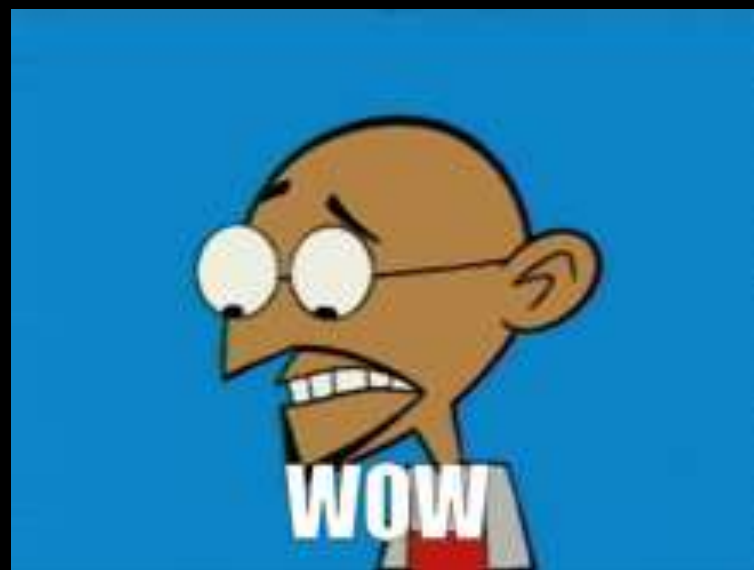


Background



UART boot





Background

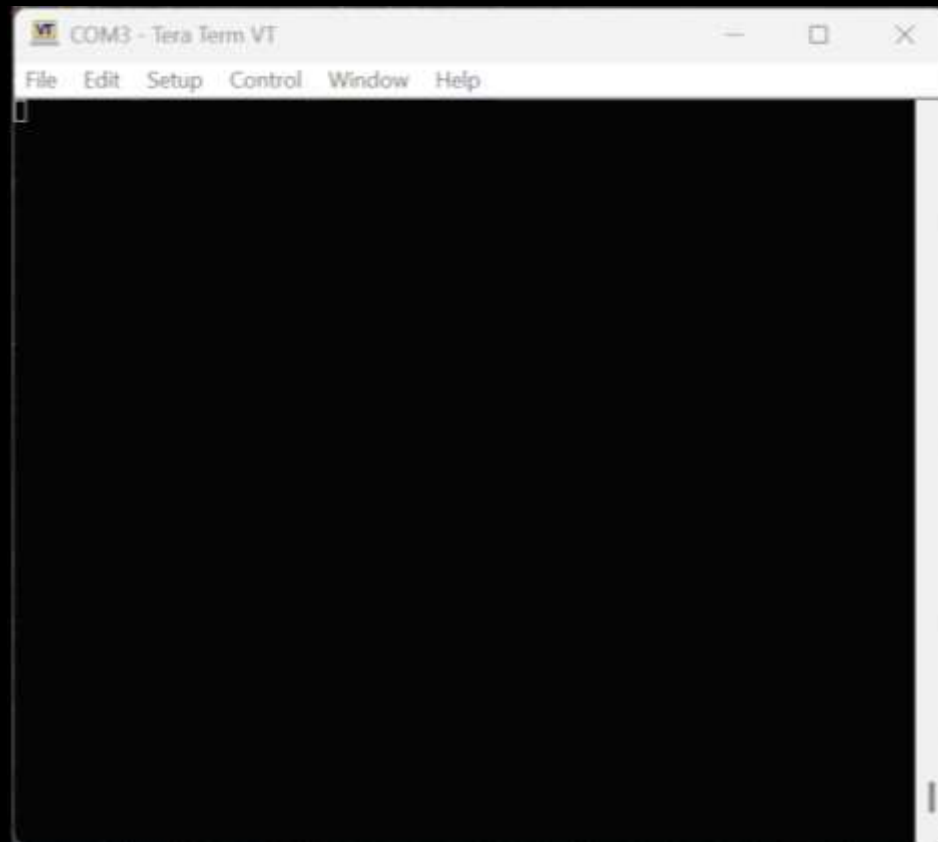
```
.  
.   
.   
MZ decomp_size=0x0030b000  
decomp_size=0x00000000  
Booting Linux on physical CPU 0x0  
Linux version 4.9.84 (nick@ubuntu) (gcc version 9.1.0 (GCC) ) #445 SMP PREEMPT Tue Mar 22 17:08:22 CST 2022  
CPU: ARMv7 Processor [410fc075] revision 5 (ARMv7), cr=50c5387d  
CPU: div instructions available: patching division code  
CPU: PIPT / VIPT nonaliasing data cache, VIPT aliasing instruction cache  
early_atags_to_fdt() success  
OF: fdt:Machine model: INFINITY6E SSC012B-S01A  
  
.   
.   
.   
disable sensor  
FW VERSION:CMK-HD510-OT1917-FW-4.6.2,builddate:Jun 7 2022,20:57:26  
  
(none) login: aision  
login: can't chdir to home directory '/home/aision'  
login[496]: root login on 'ttyS0'  
/ #
```



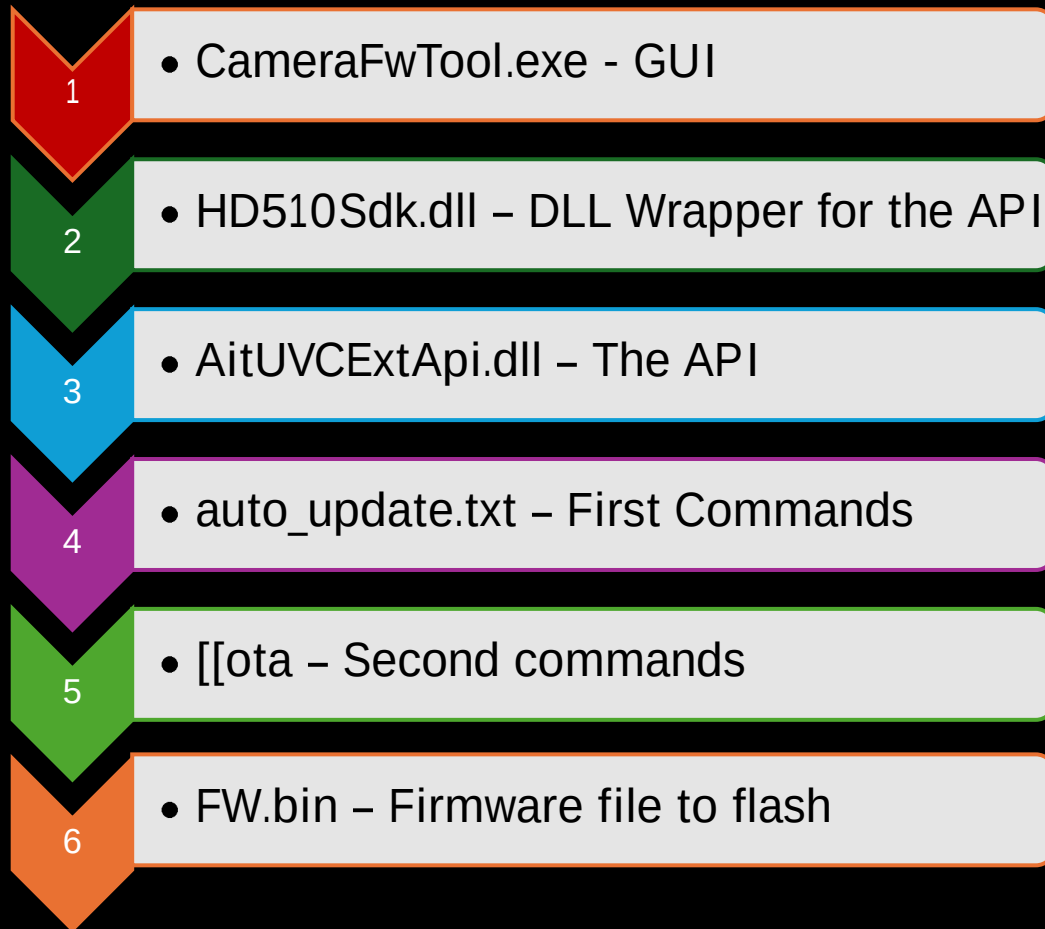


FW Update from UART

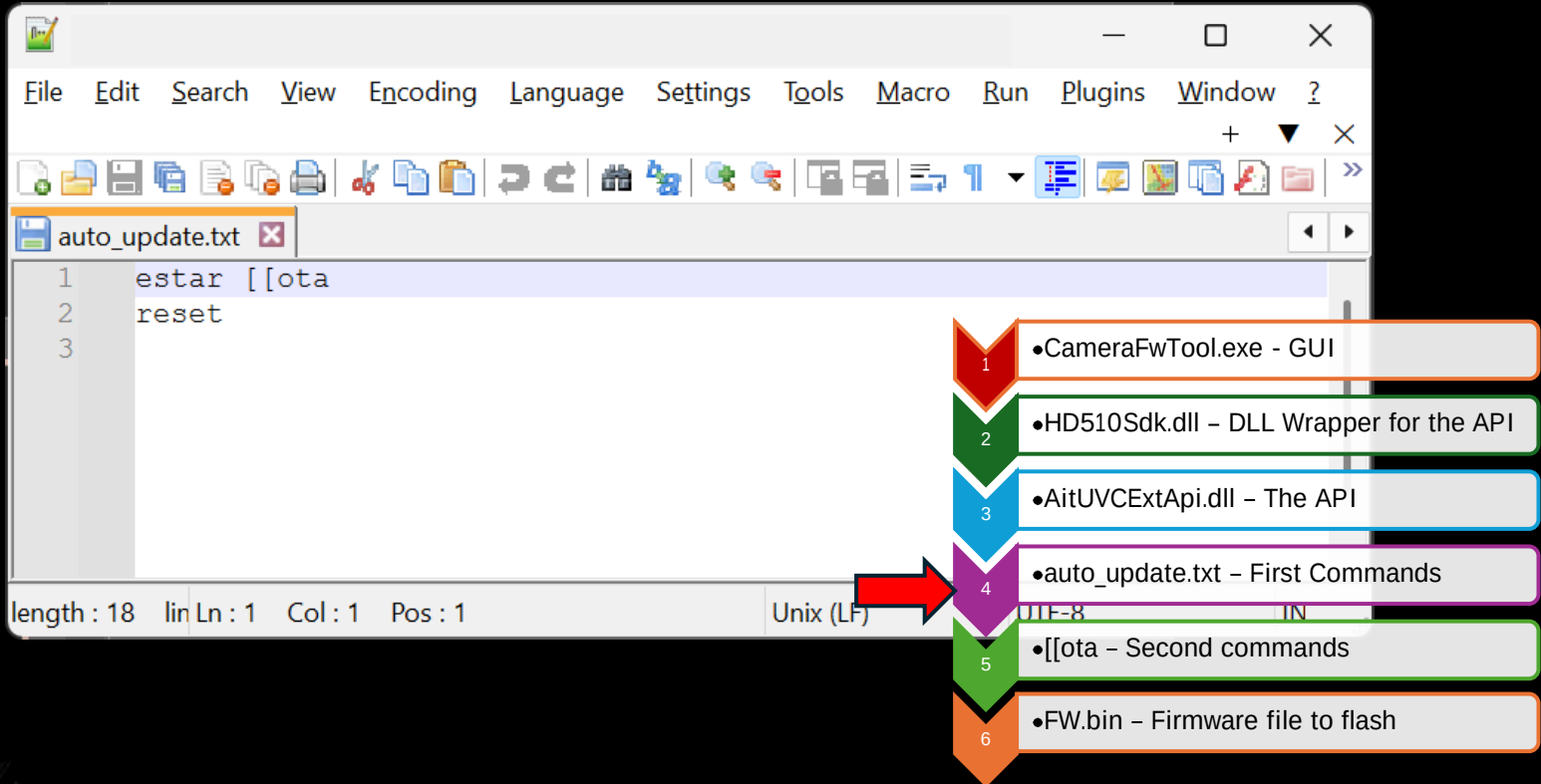
- Firmware update flow from inside the device:



Update tool flow



Back to the update tool



The screenshot shows a text editor window with the file `auto_update.txt` open. The file contains the following text:

```
1  estar [[ota
2  reset
3
```

Below the text editor, a red arrow points to a vertical list of six components, each with a numbered arrow pointing to the corresponding line in the file:

- 1 •CameraFwTool.exe - GUI
- 2 •HD510Sdk.dll - DLL Wrapper for the API
- 3 •AitUVCExtApi.dll - The API
- 4 •auto_update.txt - First Commands
- 5 •[[ota - Second commands
- 6 •FW.bin - Firmware file to flash



Back to the update tool

```
1 # <- this is for comment / total file size must be less than 4KB
2 sf probe 0
3 sf erase 0x50000 0x7B0000
4 tftp 0x21000000 lenovo_hd510_ota_v4.6.2|.bin
5 sf write 0x21000000 0x50000 0x7B0000
6 % <- this is end of file symbol
7
```

length : 215 line Ln : 4 Col : 40 Pos : 142 Unix (LF)

- 1 •CameraFwTool.exe - GUI
- 2 •HD510Sdk.dll - DLL Wrapper for the API
- 3 •AitUVCExtApi.dll - The API
- 4 •auto_update.txt - First Commands
- 5 •[[ota - Second commands
- 6 •FW.bin - Firmware file to flash



Back to the update tool

```
sf probe 0
sf erase 0x50000 0x7B0000
tftp 0x21000000 lenovo hd510 ota v4.6.2.bin
sf write 0x21000000 0x50000 0x7B0000

(UFU runcmd) sf probe 0
Flash is detected (0x070C, 0x1C, 0x70, 0x7)
SF: Detected nor0 with total size 8 MiB
(UFU runcmd) sf erase 0x50000 0x7B0000
_spi_flash_erase: addr 0x50000, len 0x7B0000 100%(cost 23063 ms)
SF: 8060928 bytes @ 0x50000 Erased: OK
(UFU loadinfo) addr 0x21000000, size 7995392
md5sum pass
(UFU runcmd) sf write 0x21000000 0x50000 0x7B0000
_spi_flash_write to 0x50000, len 0x7B0000 from 0x21000000 100%(cost 13439 ms)
SF: 8060928 bytes @ 0x50000 Written: OK
(UFU runcmd) reset
resetting ...
```

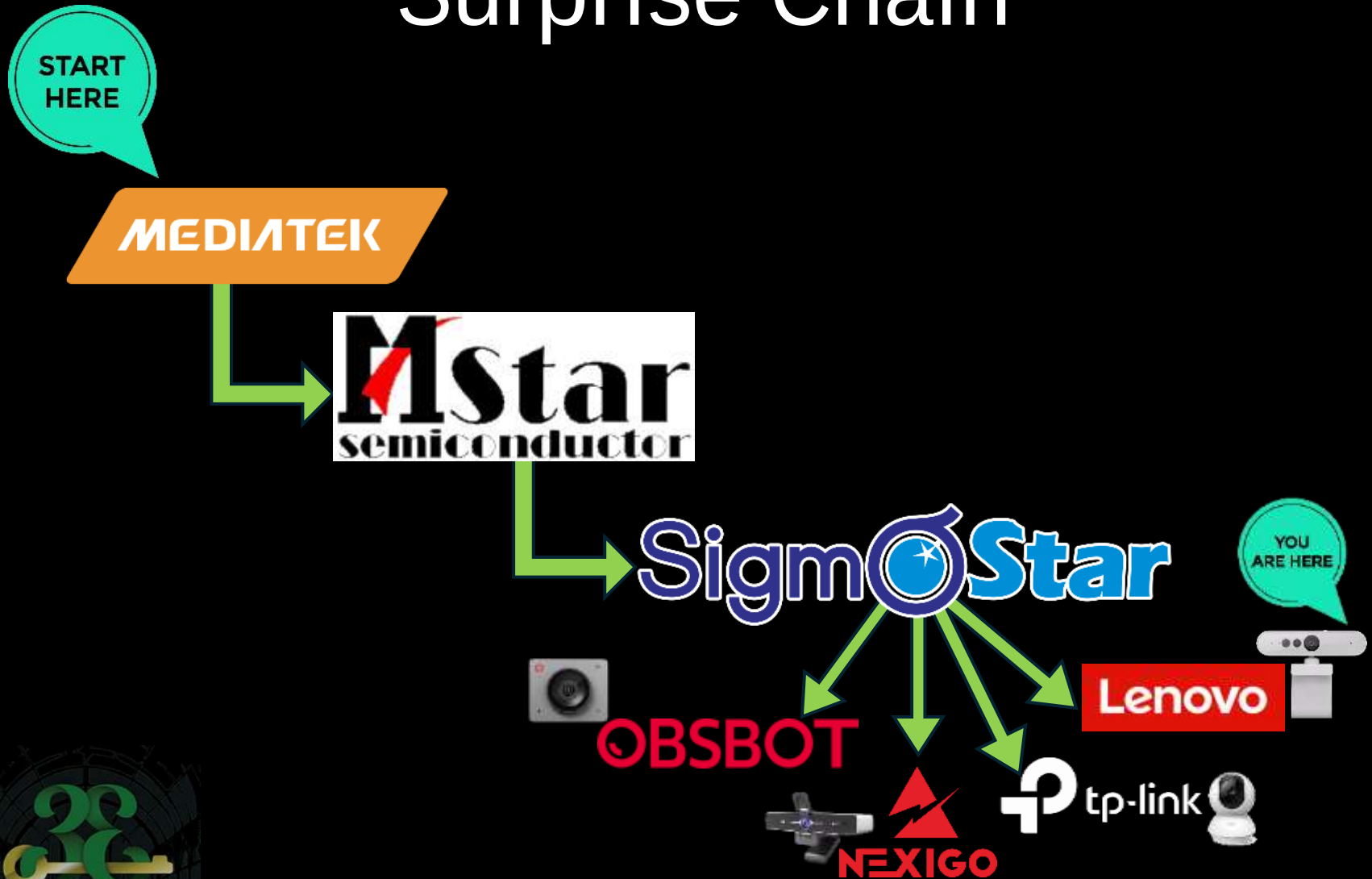


Fw update process

- We write a C# tool to flash firmware
- We share everything on GitHub/MEGA
- Links to follow



Surprise Chain



Surprise Chain

- SigmaStar websites have documentation
 - https://www.comake.online/3g/index.php?p=products_list&lanmu=2&c_id=11
- Pros:
 - Detail step by step build and toolchain setup
 - Explain the entirety of all the features in the SoC
 - Explain in detail firmware structure
 - Everything you'll every need to use the SoC
 - Shows hyperlinks to all requires files/sdk/downloads
- Cons:
 - Require authentication to access



Surprise Chain

- Trying to get more info about anything



- You have to do everything yourself
 - Google
 - GitHub
 - Forums
 - LinkedIn



Surprise Chain

- Explore the forums and websites
 - Find references to files and toolchains
 - Google for those files
 - Search for them on GitHub
 - Repeat previous steps with slightly modified names
 - ...
 - Profit
 - Bonus points – Russian Forums



Surprise Chain

- <https://github.com/DongshanPI>
 - <https://github.com/DongshanPI/SigmaStar-USBDownloadTool>
 - https://github.com/DongshanPI/Buildroot_SigmastarOriginalSDK
 - <https://github.com/DongshanPI/Sigmastar-Linux>
 - <https://github.com/DongshanPI/Sigmastar-Boot>
- <http://code.moobox.cn:8002/>
- <http://www.anjvision.com:8021/>
- <https://github.com/OpenIPC>



Surprise Chain

- No help from SoC vendors
 - TBD – There might be something by time this is presented on stage.
- So we ask Lenovo for the GPL code
 - Thanks RFP!
- Get Linux + u-boot sources
 - Build
 - Build fails, missing code
 - Request updated GPL



Surprise Chain

- We have the U-Boot and the Kernel , but we don't have the latest SDK.
- We have a collection of files from GitHub, what can we do?



Digging deep

- Firmware image breakdown (SPI-NOR)

0x0	IPL
0x1000	IPL Customer
0x2000	MXPT
0x30000	UBOOT
0x4F000	UBOOT_Env
0x50000	Kernel
0x210000	RootFS
0x4C0000	Miservice (SquashFS)
0x770000	Customer (JFFS2)



Exploitation and Fun

- We know the camera runs linux
- We know we can flash it
- We know how we can flash it
- We know what to put in the flash and where
- ...



DEMO 1



DEMO 2



YOLO / DEMO 3



Exploitation and Fun

- Go forth and have fun!
- Camera on Amazon (\$45-\$60)
<https://a.co/d/5P1ACtw>
 - Cheaper on eBay if you are willing to wait for shipping
- SPI flash tool (\$7 and up)
<https://hackerwarehouse.com/product/tigard>



Going beyond Lenovo

- Same SoC in use with other cameras
 - OBSBot – Confirmed newer kernel (5.1)
Still waiting on GPL
Insert Oday root here
 - Opal Tadpole – Confirmed linux
Still waiting on GPL



Going beyond Lenovo

- New age of more linux everywhere
- SoCs now using root of trust and secure boot
- Is the age of IoT low hanging fruit coming to an end?



Summary

- “we have determined that the Lenovo 500 FHD and Performance FHD were produced in early 2021 and do not support firmware validation. These web cameras have also reached End of development support (EODS), which means they no longer support security updates.”



Summary

- Lenovo has released a firmware update tool to update U-BOOT to support firmware update signature verification



Questions

